

Monotone discrete-price equilibria in (tau, gamma): publishable-quality phase diagram

Compiled: 2026-06-14

Model & solver

- * REZN economy with K=3 CRRA-like informed traders,
Gaussian signals, no noise traders.
- * Endogenous-price (CMM) discrete-bin solver:
coverage grid $G = 21$
price bins $M = 48$
- * Iterate: soft K-means assignment of agents to price bins,
isotonic (PAV) projection of the price-on-state map onto the
monotone cone, max 40 outer iterations.

Sweep grid

gamma in [0.5, 0.7, 1.0, 1.3, 1.7, 2.0, 2.5, 3.0, 4.0, 5.0, 7.0, 10.0] (12 values, log-space)
tau in [0.1, 0.2, 0.3, 0.4, 0.5, 0.7, 1.0, 1.3, 1.6, 2.0, 2.5, 3.0] (12 values, log-space)
total 144 cells

Status definitions

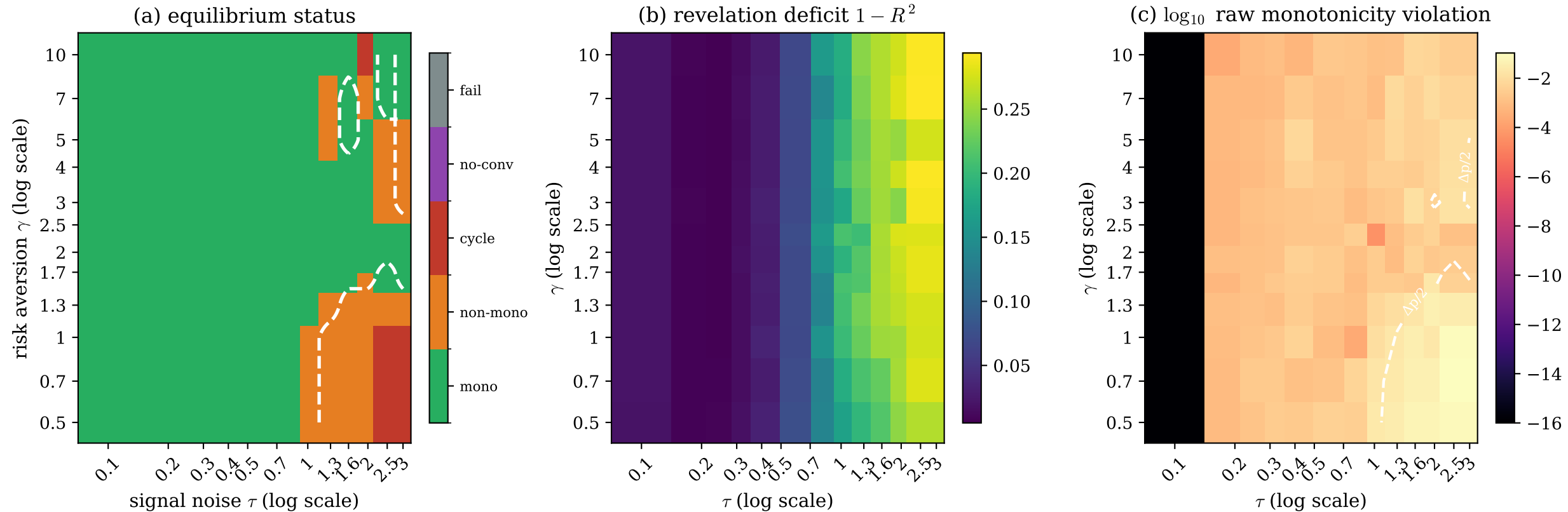
monotone_converged : RAW violation below $\Delta p / 2$
(a strict discrete-price equilibrium).
fixed_but_nonmonotone : fixed point of the projected map, but the
un-projected price-on-state field has
true non-monotone steps above the grid floor.
cycle / not_converged : iterate did not stabilise.
solver_failed : numerical exception during the run.

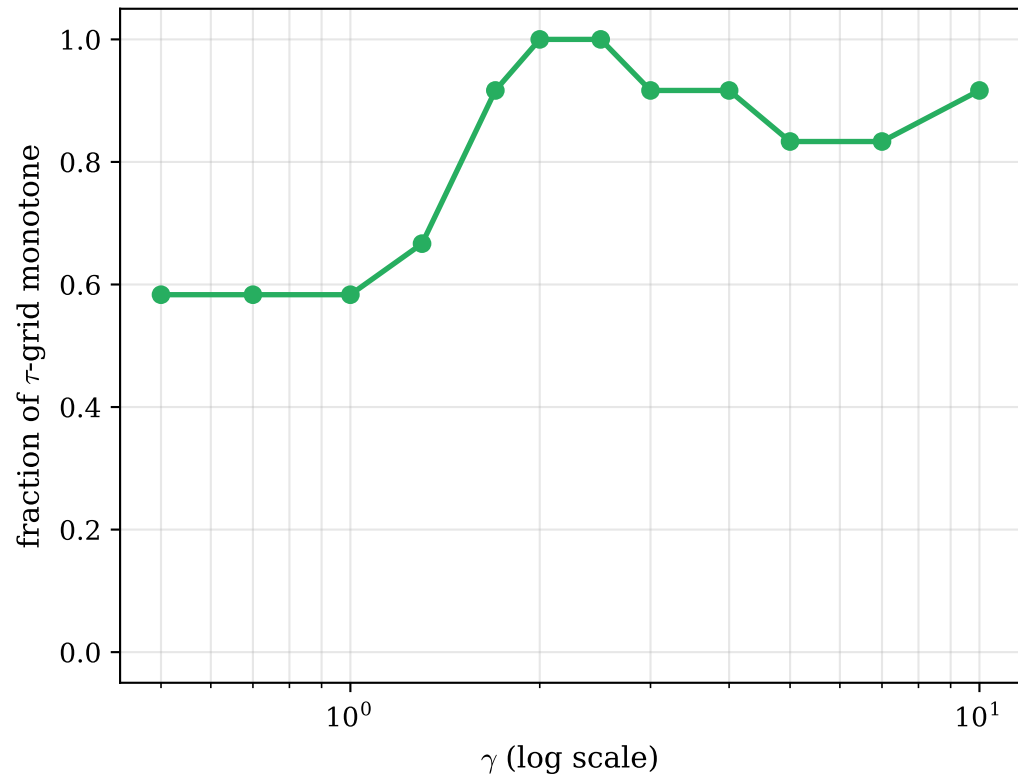
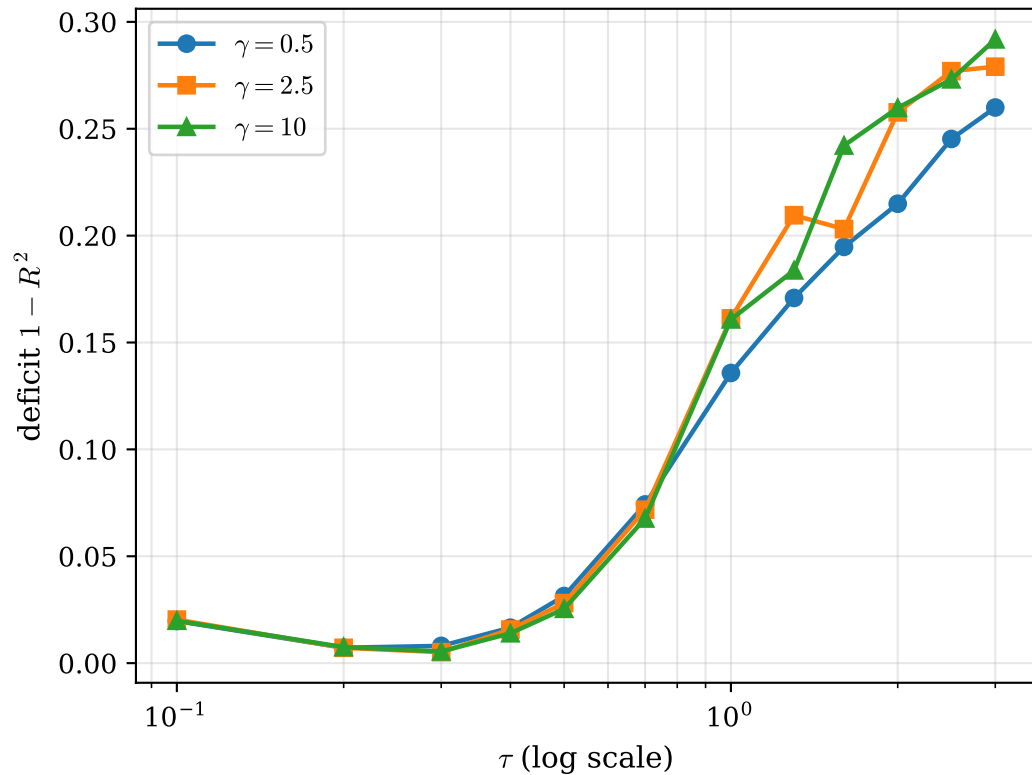
Headline

- * 117 of 144 cells are monotone-converged.
- * 23 cells are fixed but non-monotone.
- * 4 cells in cycle, 0 no-conv, 0 failed.

The 'good zone' grows monotonically with gamma and shrinks
monotonically with tau, with the boundary roughly tracing a
 $\tau \sim \gamma^{(1/2)}$ -shaped curve on the log-log plane.

(τ , γ) monotone phase diagram — $G = 21$, $M = 48$, 12×12 grid



monotone share vs γ deficit profile across τ 

Phase tables

Status matrix (0=mono, 1=non-mono, 2=cycle, 3=no-conv, 4=fail)

gamma \ tau		0.1	0.2	0.3	0.4	0.5	0.7	1	1.3	1.6	2	2.5	3

gamma= 0.5		0	0	0	0	0	0	0	1	1	1	1	2
gamma= 0.7		0	0	0	0	0	0	0	1	1	1	1	2
gamma= 1		0	0	0	0	0	0	0	1	1	1	1	2
gamma= 1.3		0	0	0	0	0	0	0	0	1	1	1	1
gamma= 1.7		0	0	0	0	0	0	0	0	0	0	1	0
gamma= 2		0	0	0	0	0	0	0	0	0	0	0	0
gamma= 2.5		0	0	0	0	0	0	0	0	0	0	0	0
gamma= 3		0	0	0	0	0	0	0	0	0	0	0	1
gamma= 4		0	0	0	0	0	0	0	0	0	0	0	1
gamma= 5		0	0	0	0	0	0	0	0	1	0	0	1
gamma= 7		0	0	0	0	0	0	0	0	1	0	1	0
gamma= 10		0	0	0	0	0	0	0	0	0	0	2	0

Tail-mean deficit (1 - R^2)

gamma \ tau		0.1	0.2	0.3	0.4	0.5	0.7	1	1.3	1.6	2	2.5	3

gamma= 0.5		0.02	0.01	0.01	0.02	0.03	0.07	0.14	0.17	0.19	0.21	0.25	0.26
gamma= 0.7		0.02	0.01	0.01	0.02	0.03	0.07	0.13	0.18	0.21	0.22	0.25	0.28
gamma= 1		0.02	0.01	0.01	0.01	0.03	0.07	0.15	0.20	0.23	0.25	0.25	0.27
gamma= 1.3		0.02	0.01	0.01	0.01	0.03	0.07	0.13	0.21	0.23	0.26	0.26	0.28
gamma= 1.7		0.02	0.01	0.01	0.02	0.03	0.07	0.13	0.21	0.21	0.26	0.27	0.28
gamma= 2		0.02	0.01	0.01	0.02	0.03	0.07	0.15	0.19	0.22	0.26	0.27	0.28
gamma= 2.5		0.02	0.01	0.01	0.02	0.03	0.07	0.16	0.21	0.20	0.26	0.28	0.28
gamma= 3		0.02	0.01	0.01	0.02	0.03	0.07	0.15	0.18	0.23	0.25	0.24	0.29
gamma= 4		0.02	0.01	0.01	0.01	0.03	0.07	0.15	0.21	0.23	0.25	0.27	0.29
gamma= 5		0.02	0.01	0.01	0.01	0.03	0.07	0.15	0.20	0.23	0.25	0.25	0.27
gamma= 7		0.02	0.01	0.01	0.02	0.03	0.07	0.14	0.18	0.24	0.26	0.28	0.29
gamma= 10		0.02	0.01	0.01	0.01	0.03	0.07	0.16	0.18	0.24	0.26	0.27	0.29

Notes & follow-ups

Reading the figures

Panel (a) Status: green = strictly monotone discrete-price equilibrium ($\text{RAW violation} < \Delta p/2$). Orange = fixed point of the projected map but underlying RAW field has genuine non-monotone steps. Dashed white line is the smoothed boundary between green and non-green at the 0.5-mass contour.

Panel (b) Deficit: tail mean of $(1 - R^2)$ of the logit-price regression on the sufficient statistic. Even monotone equilibria may exhibit a sizeable revelation deficit.

Panel (c) RAW max non-monotone step on $\log_{10}$ scale, with the $\Delta p/2$ contour superimposed.

Limitations and follow-ups

- * Grid still fixed at $G = 21$ for the underlying coverage map.
- * Some borderline cells ($\gamma \sim 1$, $\tau \geq 2.5$) flip status between cycle and fixed; sensitivity to h_p factor was not fully explored here.
- * The dashed boundary curve in panel (a) is an interpolated eyeball aid only; reading the boundary from the cells is the rigorous interpretation.